

Sai Prakash Ravilisetty

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Experience

Data Scientist

Charles Schwab

Dallas, TX, USA

09/2024 - Present

- Developed and deployed a scalable customer churn prediction model using Python, R, PySpark, and machine learning techniques, reducing customer attrition by 25% through targeted retention strategies.
- Engineered an LLM agent to automate portfolio customer inquiries, slashing manual intervention by 40% and significantly enhancing response accuracy.
- Designed and maintained an optimized data pipeline using Apache Spark and Kafka, enabling real-time data processing and improving data ingestion efficiency by 35%.
- Led the development of ML workflows from data preprocessing to model deployment, integrating CI/CD tools like Jenkins, Docker and GitLab to streamline model updates and reduce deployment time by 50%.

Data Science Intern

Copart

Dallas, TX, USA

01/2024 - 09/2024

- Led the development and deployment of an LLM Agent utilizing the GPT-3.5 model, GCP Data Catalog, BigQuery, and Langchain, achieving a 70% increase in the accuracy of automated customer support responses.
- Designed and developed an enterprise-grade application pipeline for Anomaly and Forecast of all timeseries metrics using BigQuery, Python, and Apache Airflow, reducing business response time to 60 minutes.
- Built and deployed a regression model using XGBoost, Python, and scikit-learn for predicting auction sale price of cars with 90% accuracy, resulting in a 25% increase in sales revenue.
- Crafted metric-driven reports with Python and SQL, delivering insights resulting in a 15% increase in operational efficiency across departments.

Software Engineer, Machine Learning

Verizon

Hyderabad, India

02/2020 - 08/2022

- Awarded "Verizon Spotlight Award" for developing a machine learning analytical model and recommendation engine to recommend products to segmented customers and identified key trends and patterns using scikit-learn, pandas, and NumPy, driving \$1M in revenue.
- Designed and developed an LSTM Model using Keras, Python, and TensorFlow to perform classification and sentiment analysis, predicting the topic and degree of negativity of unstructured customer reviews, driving 95% customer support enhancement.
- Developed and deployed an Optical Character Recognition model using OpenCV, Python, and East OCR model, extracting text from images with 89% accuracy and reducing inconsistencies by 25%.
- Created and maintained data-driven dashboards for customer behavior using SQL and Tableau, improving customer retention by 80%.

Data Science Intern

Flipkart

Bengaluru, India

05/2019 - 07/2019

- Engineered and implemented a BERT segmentation model using PyTorch, Python, and Hugging Face Transformers, reducing defect resolution time by 90% and enabling proactive resolution.
- Conducted A/B testing using Python to evaluate and optimize algorithm effectiveness, achieving a 30% improvement in performance.

Skills

- Programming Languages:** Python, R, PySpark, SQL, NoSQL, MongoDB, Chroma DB, BigQuery
- Machine Learning:** Regression, Classification, Clustering, Neural Networks, CNN, R-CNN, RNN, LSTM, BERT, LLM, Langchain, Llama, Transformers, Sentiment Analysis, Forecasting, Anomaly Detection, Recommendation System, Image Processing, Image Segmentation
- CI/CD Tools:** GIT, Linux, Jenkins, Docker, Postman, GitLab, GitHub
- Data Science Tools:** TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, SciPy, OpenCV, Tableau, PowerBI
- Cloud platforms:** Amazon Web Services (AWS), Google Cloud Platform (GCP)
- Data Engineering:** Hadoop, Apache Spark, Kafka, Hive, Apache Airflow, DBT, Pentaho
- Certifications:** Google Cloud Machine Learning Engineer, AWS Solution Architect Associate, AWS Cloud Practitioner, Deep Learning Specialization, Problem Solving HackerRank, SQL Advanced HackerRank

Education

- Master of Science in Information Technology and Management** | University of Texas at Dallas, Dallas, TX, USA 08/2022 - 05/2024
- Bachelor of Science in Computer Science and Engineering** | Gitam University, Visakhapatnam, India 07/2016 - 05/2020

Projects

- Plant Pathology Disease Detection:** Developed an Ensemble R-CNN plant pathology disease detection system employing image segmentation and object detection using Python and PyTorch, achieving 91% test accuracy.
- Stack Overflow Question Tag Classification:** Developed a Transformer multi-label classification model using PyTorch, Python, and Hugging Face Transformers for Stack Exchange tags with 94% labeling accuracy.
- ANT Truck Accident Analysis:** Designed and deployed a data pipeline using HDFS, MapReduce, Hive, and Flume to process large-scale ANT trucking data, uncovering critical risk factors for accidents, which reduced potential hazards by 15%.
- Live Facial Recognition:** Spearheaded a team in developing a custom CNN architecture for live facial recognition using Python, MTCNN and TensorFlow, achieving 97% accuracy.

Achievements

- Published a Research Journal at IRJET 2020 titled "Blindness detection based on Retinal Images", where we designed an effective algorithm leveraging color, texture, and pattern analysis on fundus photography of dilated retina.
- As an **AI Student Assistant** at UTD, I designed and developed a Vulnerability classification model and performed Bug Report Analysis to improve campus security, mitigating 20% of vulnerability and reducing reported issues by 60%.
- Awarded the prestigious **Dean's Excellence Scholarship** for outstanding academic achievement and leadership. (07/2022)
- Received the esteemed **Lion's Club Scholarship** for exceptional academic performance and community involvement. (07/2023)
- As part of **AWS Student Club**, planned and Executed AWS Milestone Day attracting 200 graduate students at UTD (11/2022)